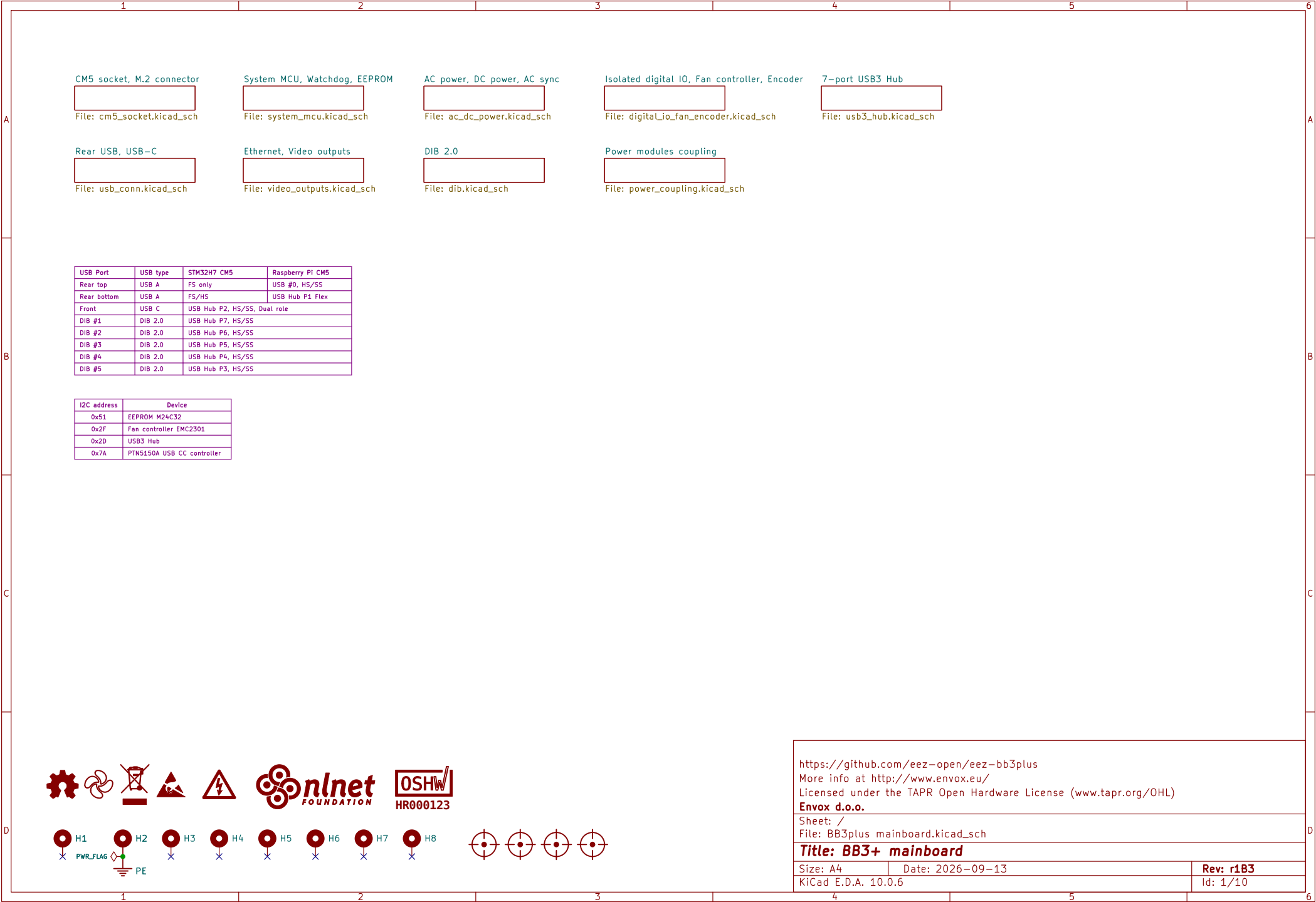
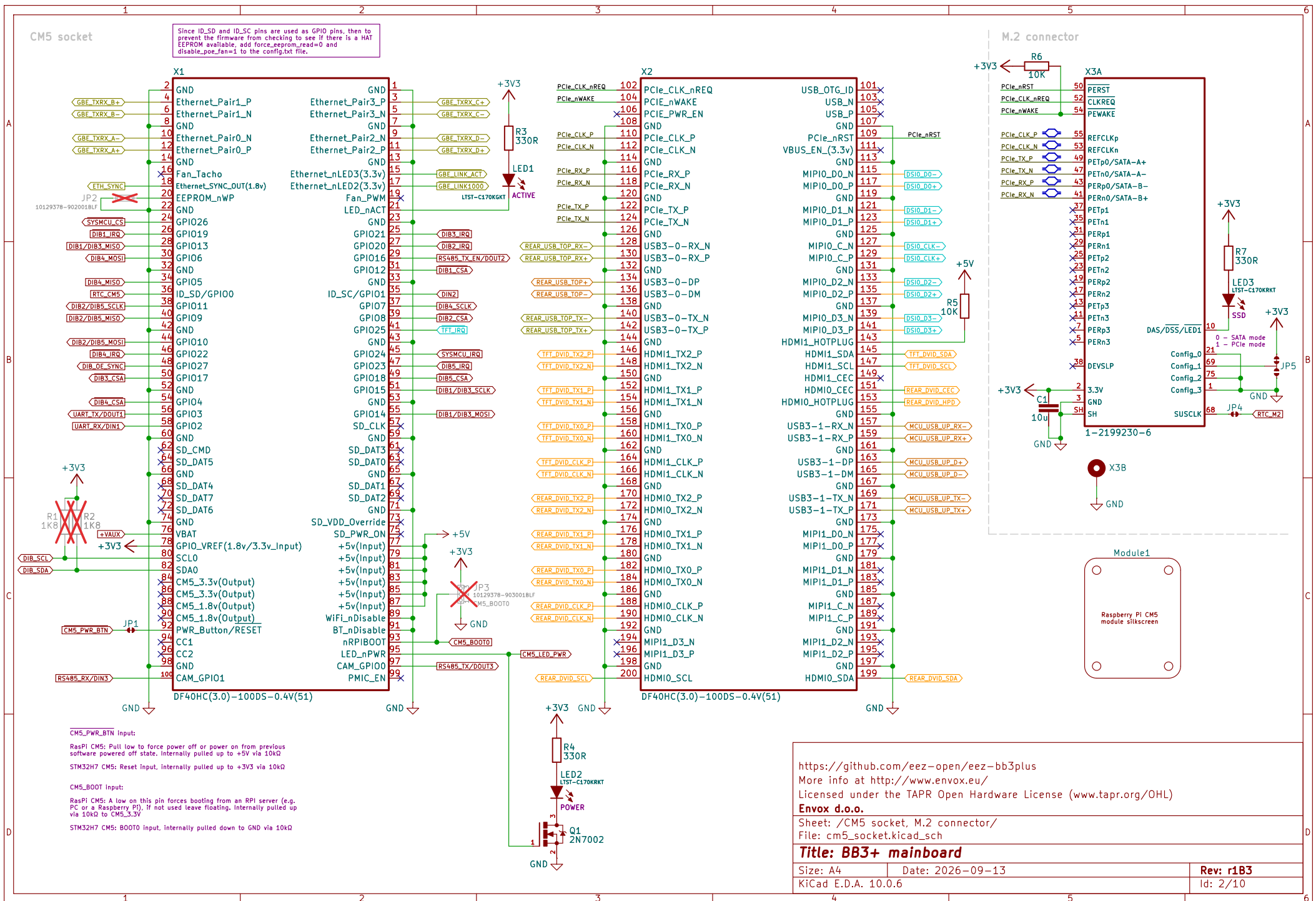
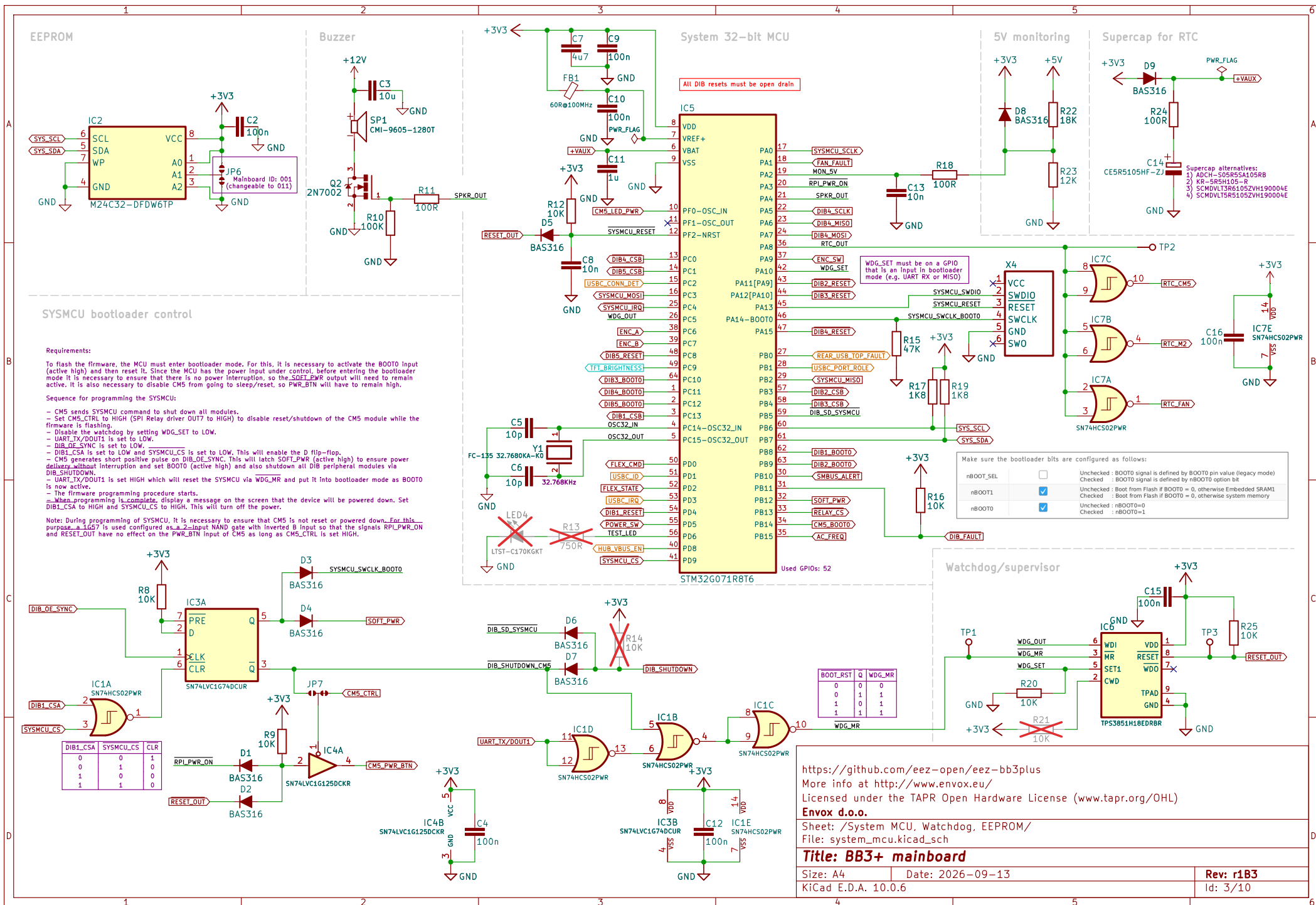
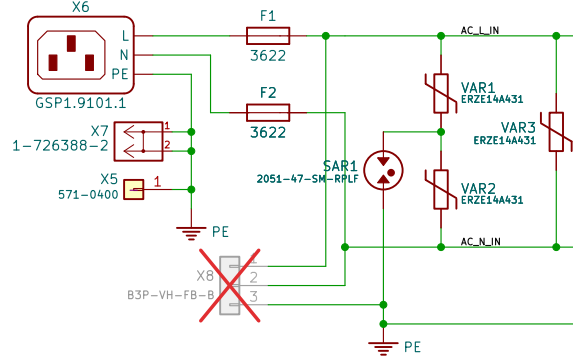




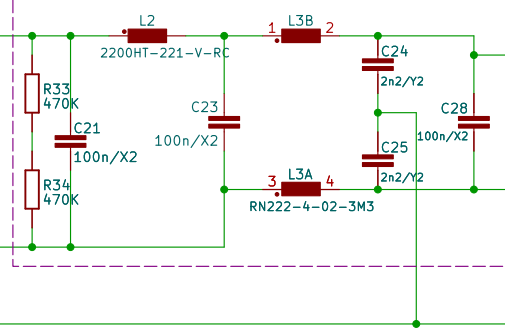




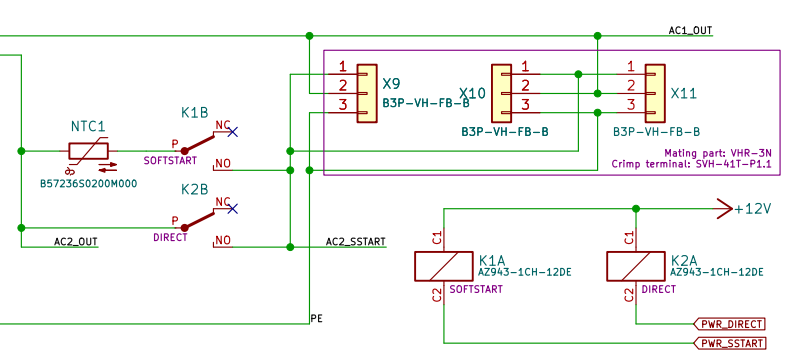
AC inlet, filtering and protection



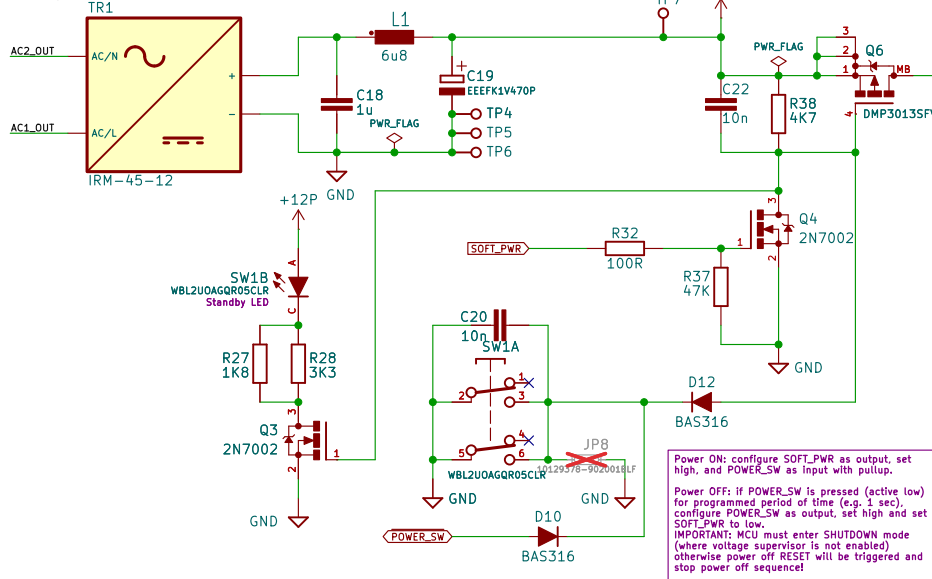
EMI compliance filter



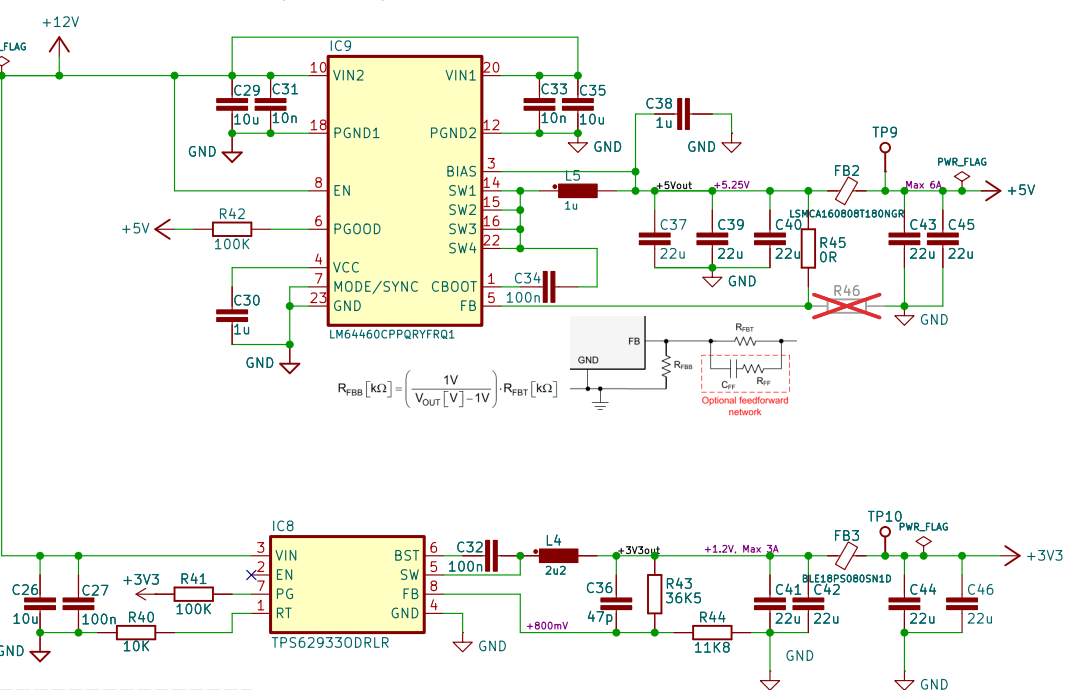
AC power control, Inrush current limiter



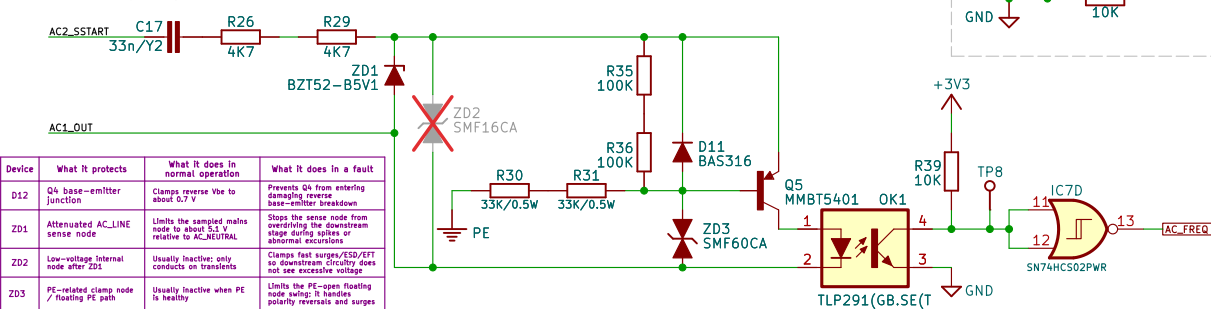
Soft power switch



DC-DC stepdown converters (+5V, +3V3)



AC Mains frequency and PE fault detection



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /AC power, DC power, AC sync/

File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

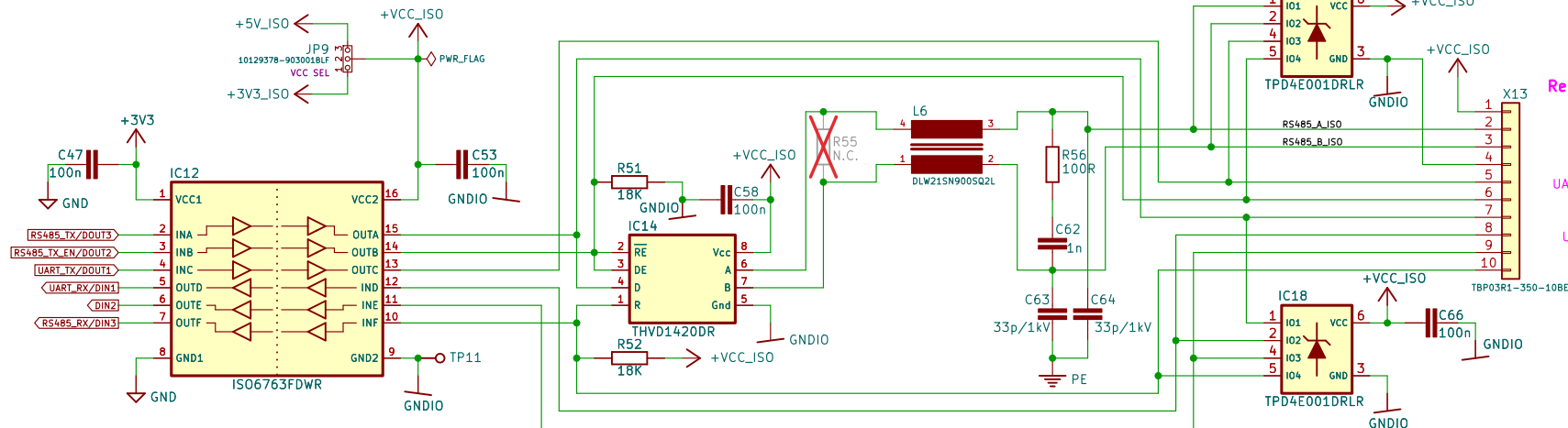
Size: A4 Date: 2026-09-13

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Id: 4/10

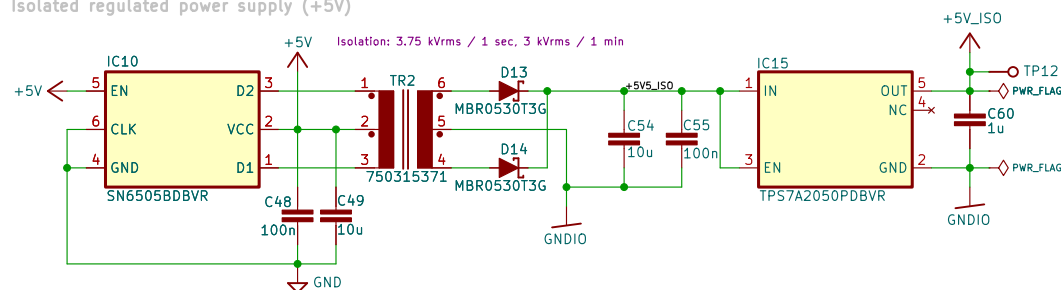
Isolated digital I/O & RS485 transceiver (half-duplex)



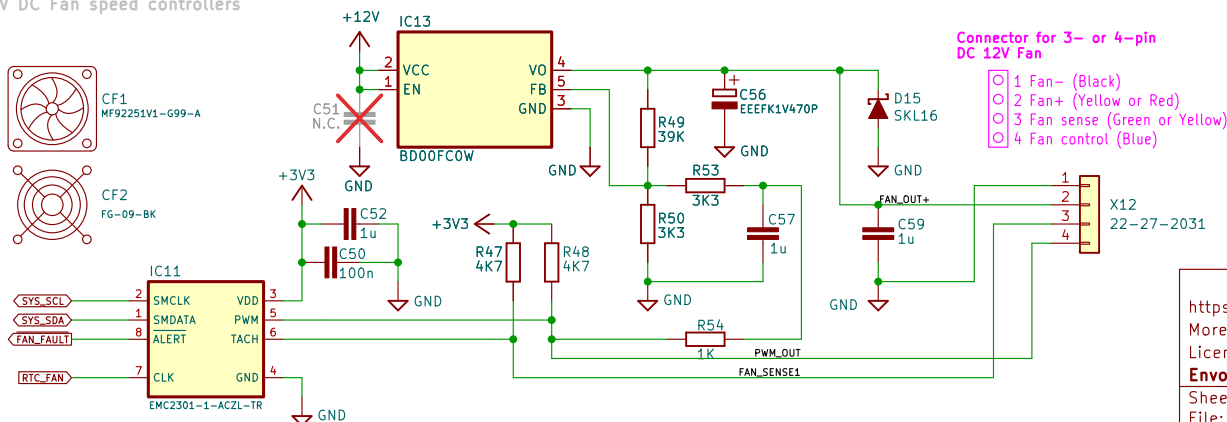
Rear panel I/O connector

+VCC_ISO	1
RS485_A_ISO	2
RS485_B_ISO	3
GND_ISO	4
UART_TX/DOUT1	5
DOUT2	6
DOUT3	7
UART_RX/DIN1	8
DIN2	9
DIN3	10

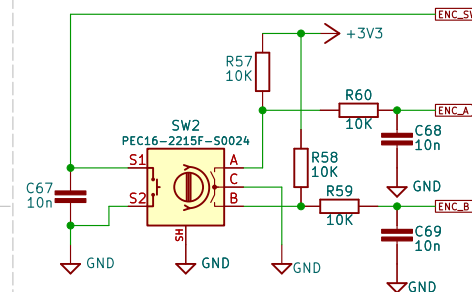
Isolated regulated power supply (+5V)



12V DC Fan speed controllers



Encoder with switch



Encoder shaft length: 15 mm / Flatted

Knob head Ø31mm A1031648 (OKW) + Knob pointer A1101007 (OKW)
Optional: Knob head Ø31mm A3131068 (OKW) + Knob cap A3231007 (OKW)
Knob head Ø24mm A1024648 (OKW) + Knob pointer A1101007 (OKW)

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

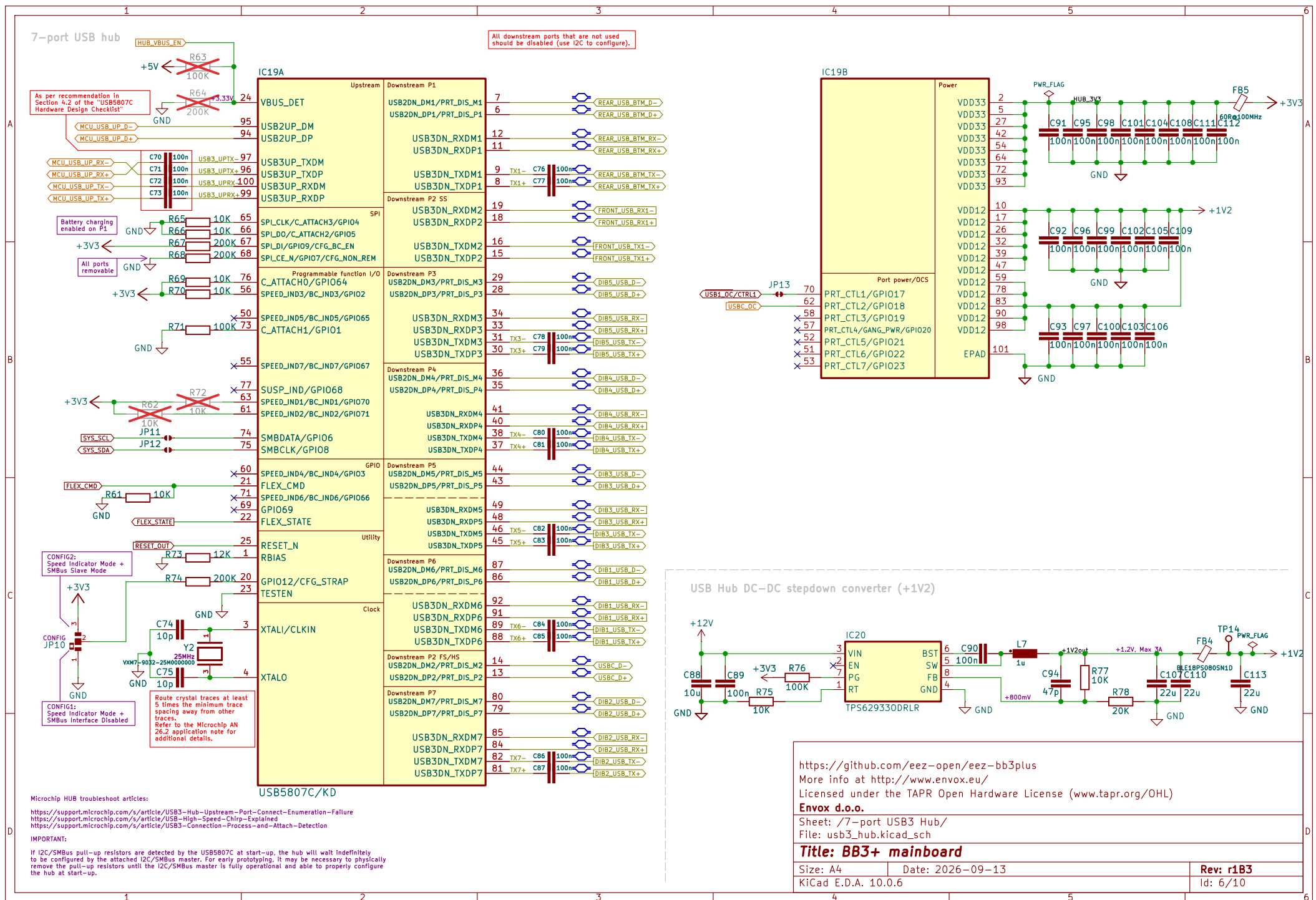
Title: BB3+ mainboard

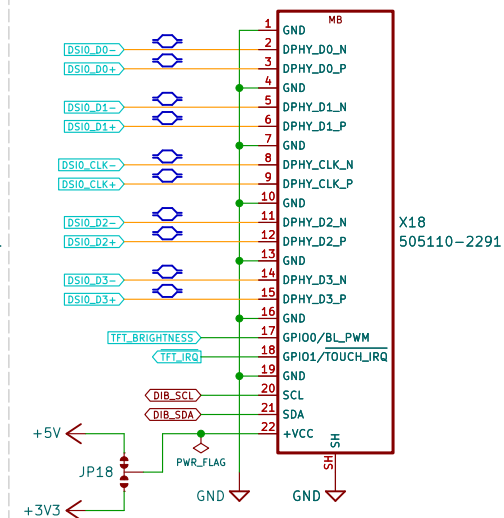
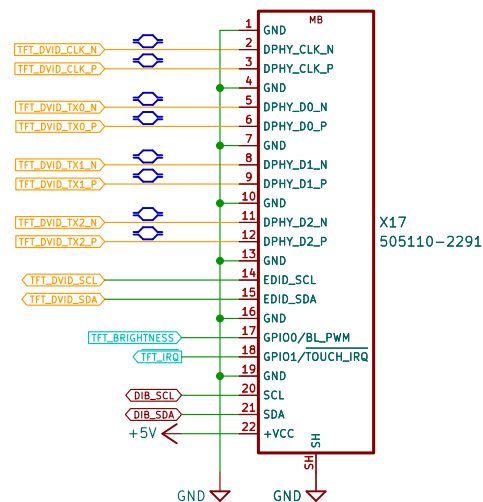
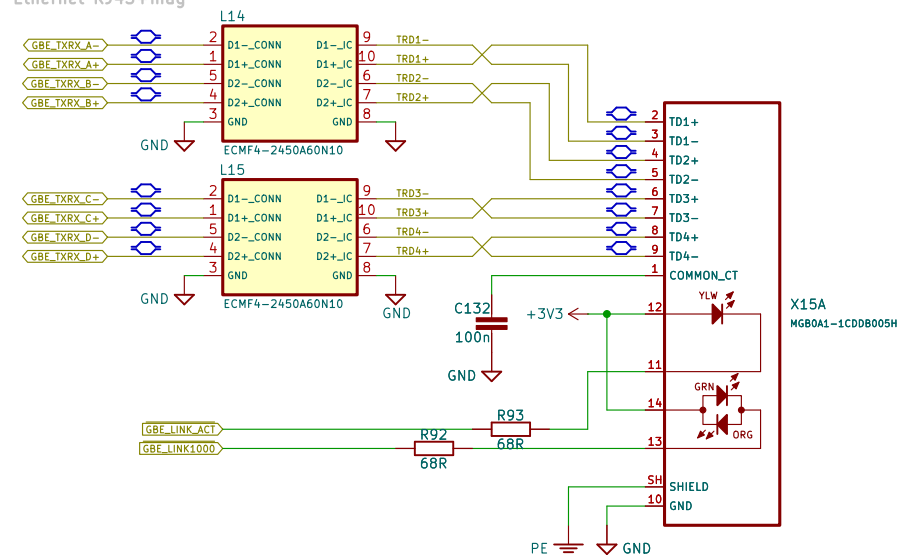
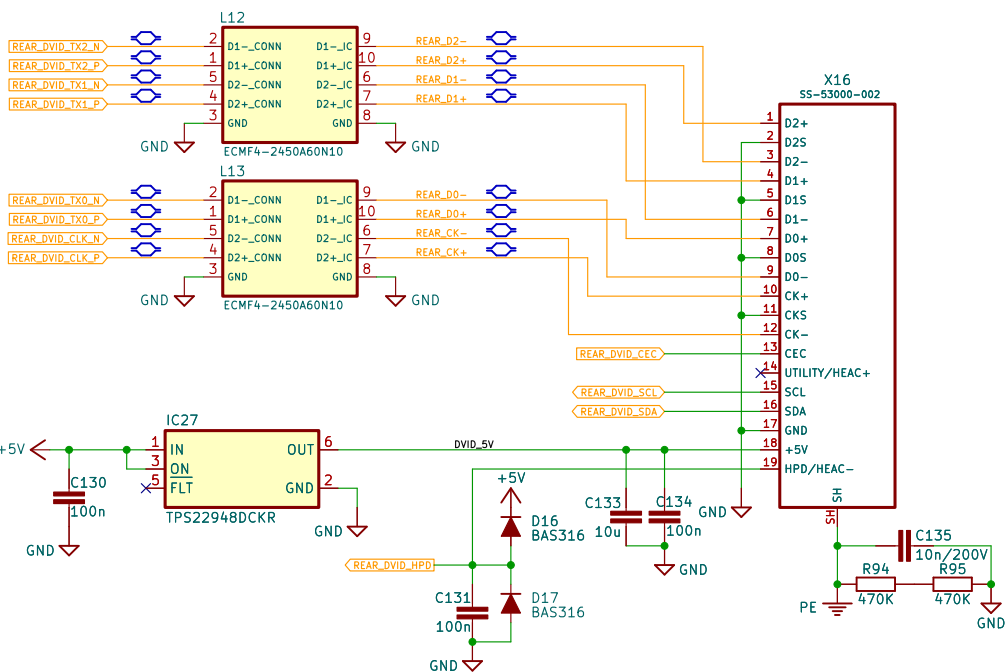
Size: A4 Date: 2026-09-13

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Id: 5/10





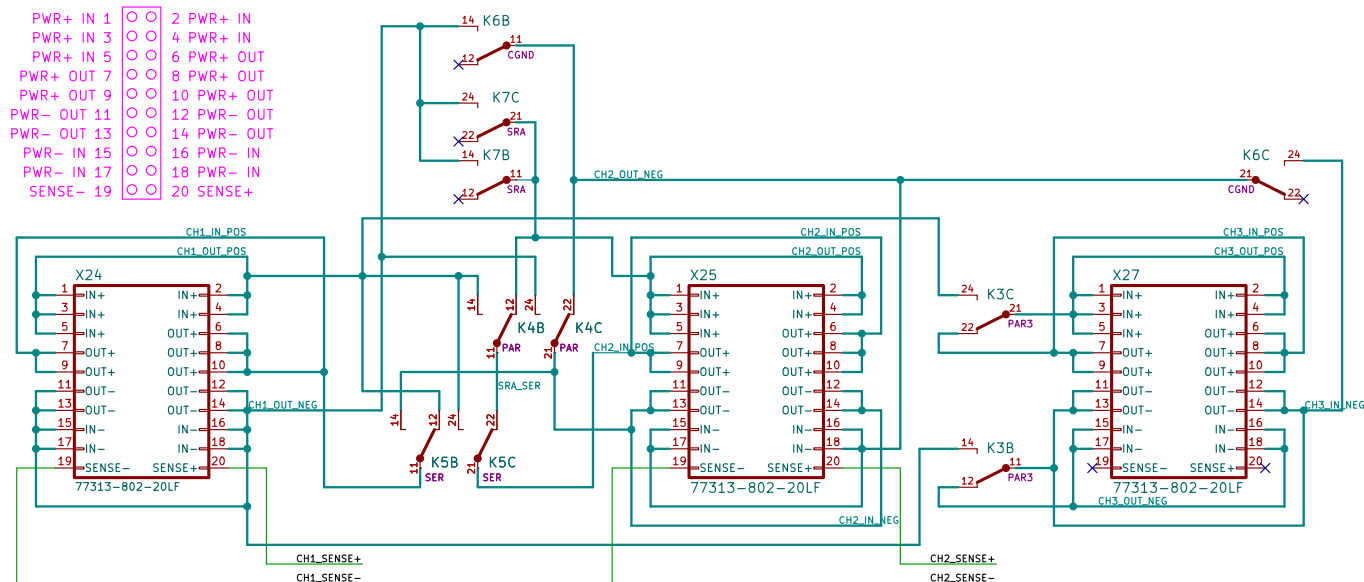
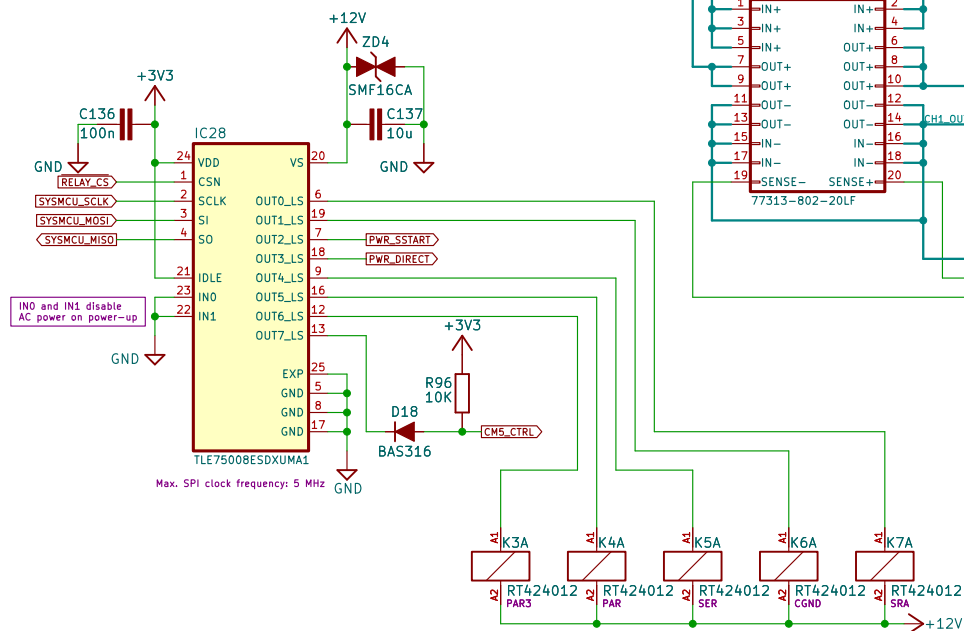
4

Id: 8/10

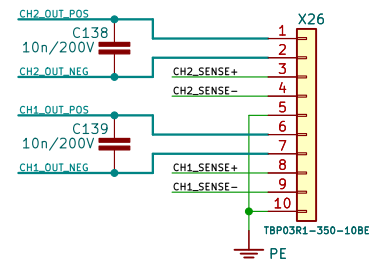
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	☐	☐	2 PWR+ IN
PWR+ IN 3	☐	☐	4 PWR+ IN
PWR+ IN 5	☐	☐	6 PWR+ OUT
PWR+ OUT 7	☐	☐	8 PWR+ OUT
PWR+ OUT 9	☐	☐	10 PWR+ OUT
PWR- OUT 11	☐	☐	12 PWR- OUT
PWR- OUT 13	☐	☐	14 PWR- OUT
PWR- IN 15	☐	☐	16 PWR- IN
PWR- IN 17	☐	☐	18 PWR- IN
SENSE- 19	☐	☐	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+	1	☐
CH2_OUT-	2	☐
CH2_SENSE+	3	☐
CH2_SENSE-	4	☐
PE	5	☐
CH1_OUT+	6	☐
CH1_OUT-	7	☐
CH1_SENSE+	8	☐
CH1_SENSE-	9	☐
PE	10	☐

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

FILE: DDB1 - 11/11/2016

Size: A4	Date: 2026-09-13
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Rev: r1B3

Id: 10/10